

# INDUSTRY 4.0

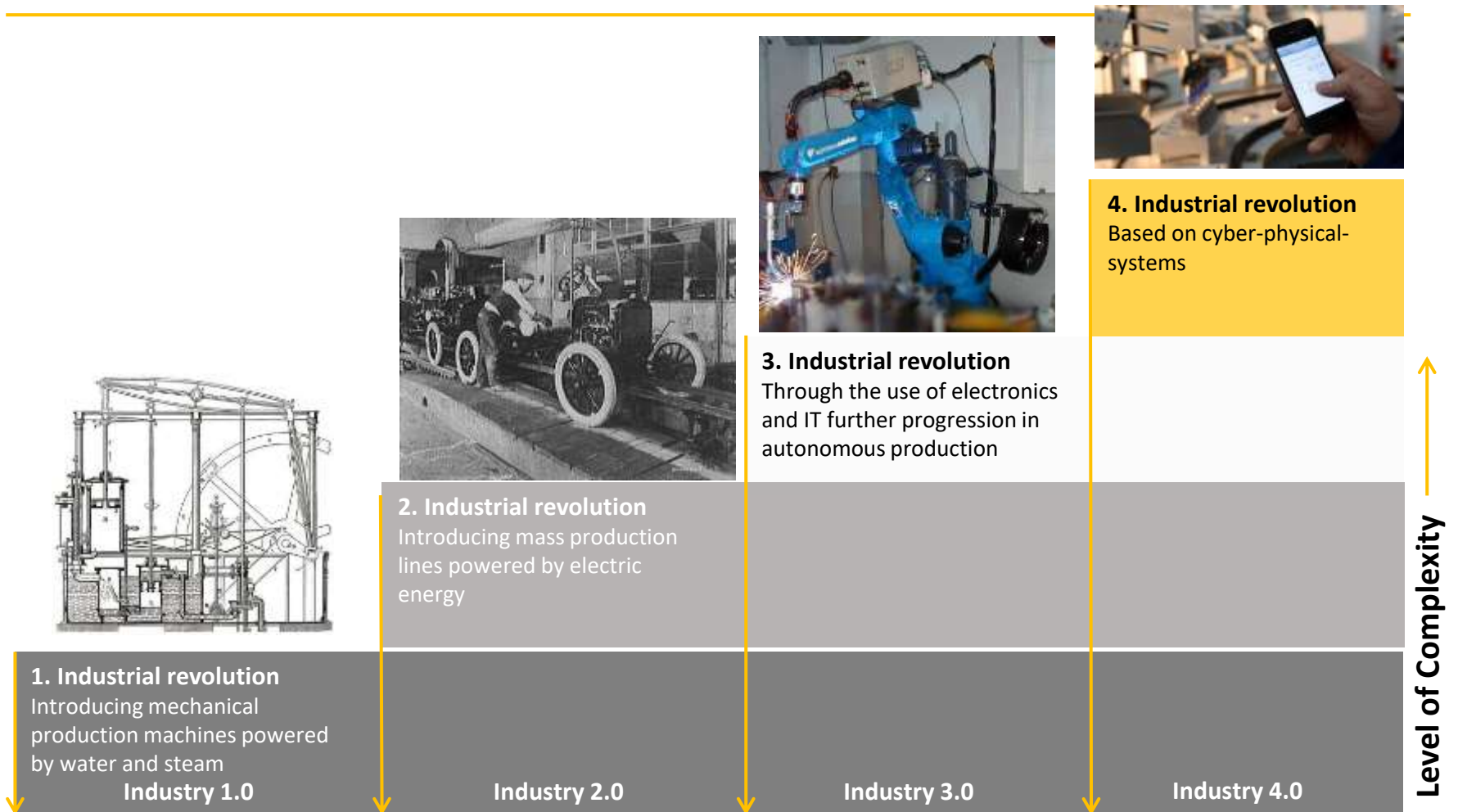


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# Industrial Evolution



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# Industry 4.0: Germany

## Smart Manufacturing Leadership Coalition: USA

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Based on the technological concepts of cyber-physical systems, the Internet of Things and the Internet of Services, it facilitates the **Vision of the Smart Factory**.

Within the modular structured Smart Factories of Industry 4.0, cyber-physical systems monitor physical processes, create a virtual copy of the physical world and make decentralized decisions.

Over the Internet of Things, Cyber-physical systems communicate & cooperate with each other & humans in real time. Via the Internet of Services, both internal & cross-organizational services are offered & utilized by participants of the value chain.

Builds on the Digital revolution  
Smaller & powerful sensors  
Machine Learning

Ubiquitous internet  
Artificial Intelligence (AI)  
Labor & Energy Cost

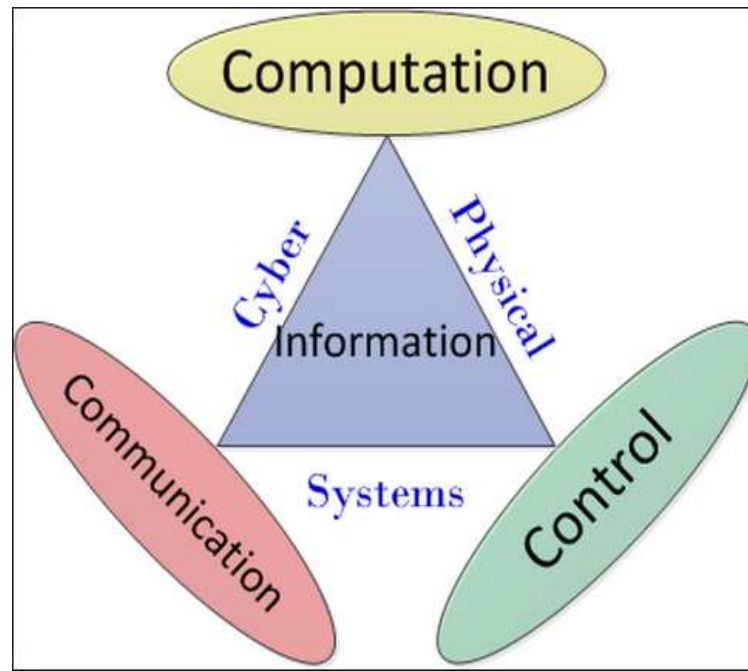


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# Cyber Physical System In Smart Manufacturing

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A **cyber-physical system (CPS)** is a system of collaborating computational elements controlling physical entities. CPS are physical and engineered systems whose operations are monitored, coordinated, controlled and integrated by a computing and communication core. They allow us to add capabilities to physical systems by merging computing and communication with physical processes.



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# Task & Challenges

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1. Generating digital data (Parameters) from Old Spot Weld & Projection Weld Machine.
2. Collection of produced data from each machine on shop floor.
3. Transportation of data from Shop floor to Office PC.
4. Creating platform (PC Application/ web portal) to view & log the data.
5. Pushing data on cloud.

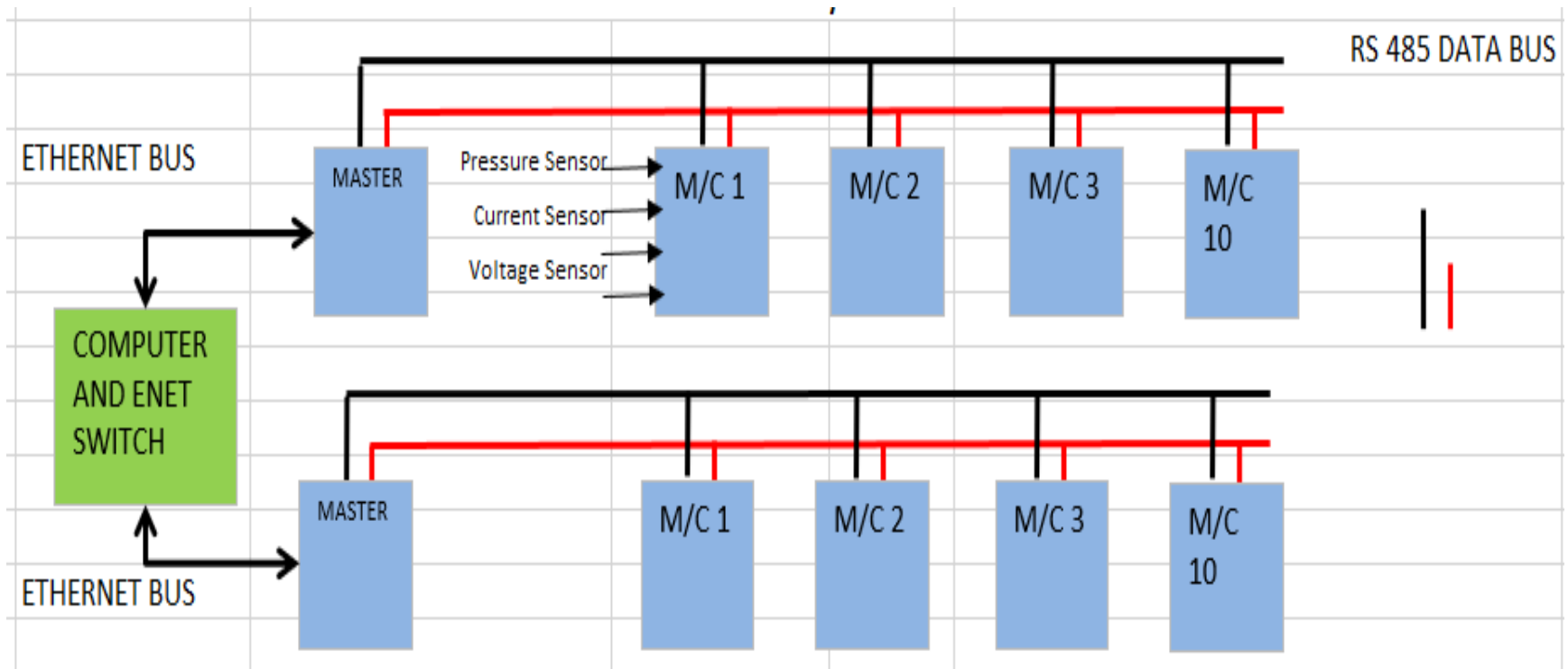


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# Block Diagram

## Smart Data Collection of Welding Shop

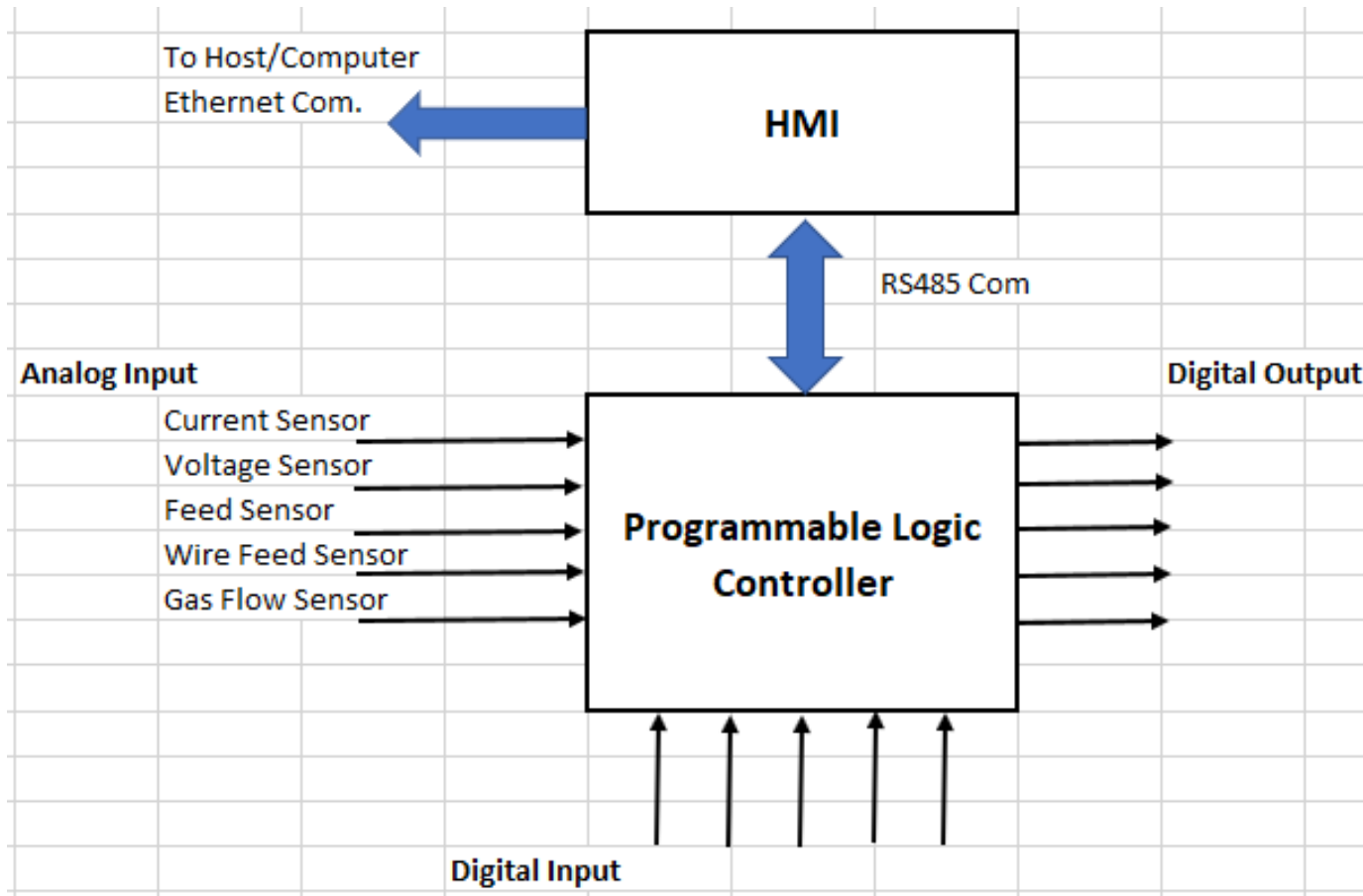


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# Block Diagram

## Each Welding Machine



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# Parameters Of Machine Spot Weld and Projection Weld

| Sr. No. | Parameter              |
|---------|------------------------|
| 1       | Operator code          |
| 2       | Work Order No.         |
| 3       | Work Order Qty         |
| 4       | Air Pressure Healthy   |
| 5       | No. of strokes         |
| 6       | Air Pressure           |
| 7       | Primary Weld Current   |
| 8       | Force by air pressure  |
| 9       | Water Flow Healthy     |
| 10      | No of job              |
| 11      | Weld Time              |
| 12      | Squeeze Time           |
| 13      | Hold Time              |
| 14      | Water Flow             |
| 15      | Force by load cell     |
| 16      | Secondary Weld current |
| 17      | Secondary Voltage      |
| 18      | Loss Code              |

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# Action Plan

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1. Selection of hardware & transducers/sensors.
2. Manufacturing of Customized sensor if unavailable.
3. Application/ mounting of sensors on machine.
4. Manufacturing control panels.
5. Programming of Controller
6. Converting Raw data to desire readable form.
7. Programming of HMI.
8. Pushing data on data bus.
9. Validating data on platform
10. Logging & analysis



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# Application Photos & Videos

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| W 01 SPOT WELD MACHINE |          |                    |         | SETTING |
|------------------------|----------|--------------------|---------|---------|
| W.O. NUMBER            | 2222     | PRESSURE (BAR)     | 5.8     |         |
| W.O. QTY.              | 4000     | PRI. WELD CURRENT  | 28.00 A |         |
| OPERATOR CODE          | 7226     | FORCE BY PRESSURE  | 478 KG  |         |
| NO.OF STROKE           | 737      | NO.OF JOB          | 4245    |         |
| WELD TIME (MS)         | 41       | SQUE. TIME (MS)    | 20      |         |
| HOLD TIME (MS)         | 8        | FORCE BY LOAD CELL | 101 KG  |         |
| SEC. WELD VOLTAGE      | 0.00 A   | SEC. WELD CURRENT  | 0.00 A  |         |
| WATER FLOW             | 2.79 LPM | OP. NO.            |         |         |

NEXT



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# Application Photos & Videos

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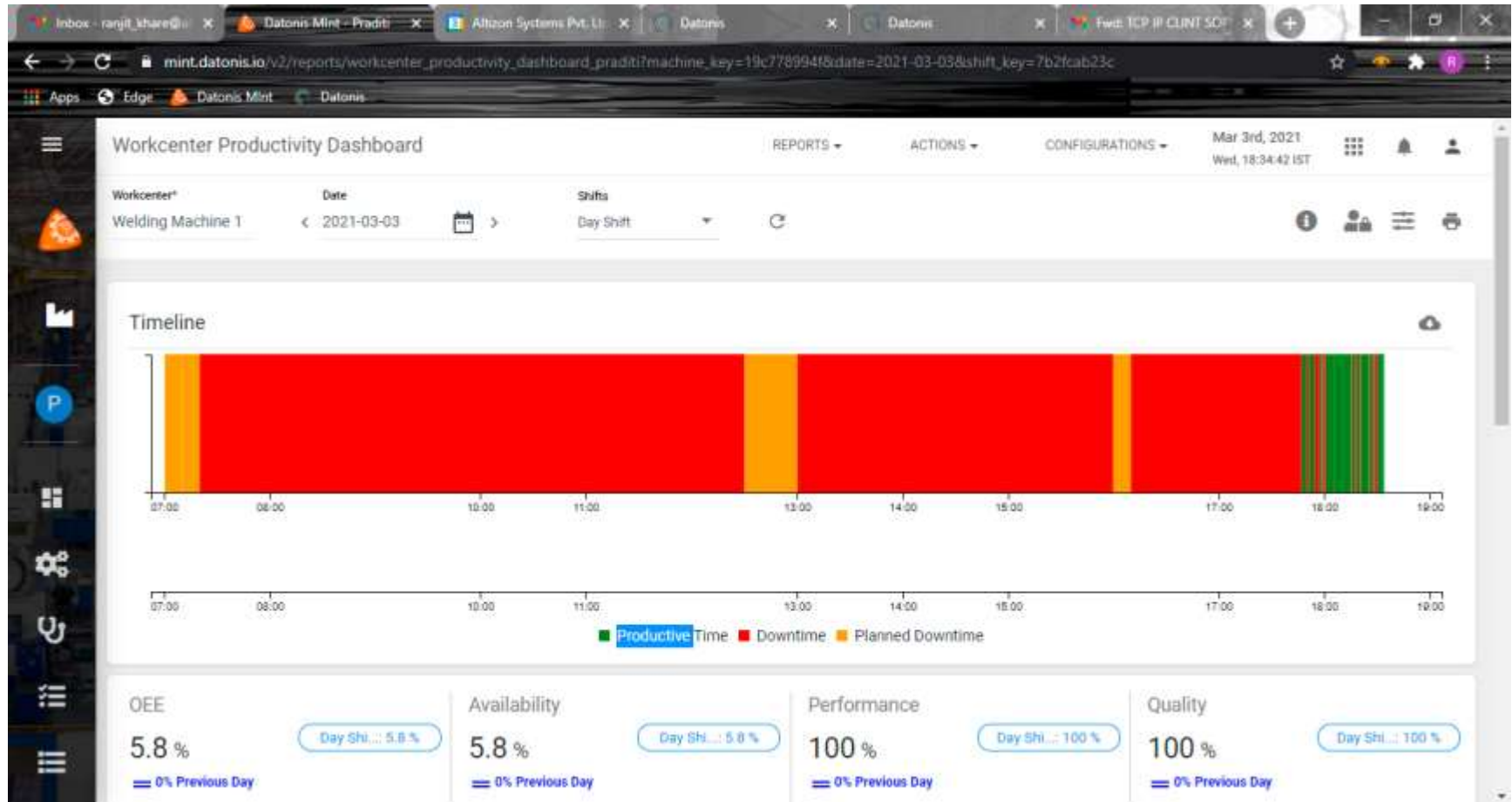


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# Application Photos & Videos Web Portal & Analysis Tool



Web platform under trial



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# Thank you



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